



Decision Support

Multi-objective decision analysis for competence-oriented project portfolio selection

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ABSTRACT

This paper develops a multi-objective optimization model for project portfolio selection taking employee competencies and their evolution into account. The objectives can include economic gains as well as gains expressed in terms of aggregated competence increments according to pre-defined profiles. In order to determine Pareto-optimal solutions, the overall problem is decomposed into a master problem addressing the portfolio selection itself, and a slave problem dealing with a suitable assignment of personnel to the work packages of the selected projects over time. We provide an asymptotic approximation of the problem by a linearized formulation, which allows an efficient and exact solution of the slave problem. For the solution of the master problem, we compare the multi-objective metaheuristics NSGA-II and P-ACO. Experimental results both for synthetically generated test instances and for real-world test instances, based on an application case from the E-Commerce Competence Center Austria, are presented.

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1. Introduction

Management is about doing the right things and making sure that things are done right. In research-centered organizations, this involves identifying the “best” portfolio of research and development (R&D) projects to be pursued out of a given set of project proposals and setting these up properly, i.e., scheduling them and providing them with the necessary resources (e.g., by assigning researchers to them). These tasks are challenging for a variety of reasons: firstly, the problem of selecting a subset of projects in the presence of limited resources constitutes a problem of knapsack type which is known to be NP-hard. Secondly, in order to assess the feasibility of a given portfolio, the projects included in it must be scheduled and staff must be assigned, two (sub-) tasks that are difficult themselves. Thirdly, staff assignment determines the further development of the employees’ competence levels, which influences their ability to work in later projects. Observe that it is exactly by implementing the selected projects that new skills are attained, contributing to the (then extended) available competence resources. Finally, the problem at hand is characterized by incommensurate and conflicting objectives. While economic objectives (e.g., return of investment) are common in literature, the explicit consideration of (additional) competence

objectives is motivated by the fact that competencies increasingly determine the attainment and sustainability of strategic positions in market competition. Since specific competencies cannot be easily found on the (job) market at any given time in the required quality and/or capacity and integrating new members in an established team of researchers may entail considerable costs in terms of communication and coordination efforts, many companies opt for a gradual internal development of competencies. However, the in-house “production” of competencies may also be fairly costly and time-consuming. The decision on which competencies to develop to which degree is therefore of high managerial relevance.

This paper draws from three strands in the project management literature, combines them in a unified model, and establishes efficient algorithmic solution techniques. The first strand is formed by articles on *project portfolio selection* that either support the choice of suitable project portfolios by means of various approaches (e.g., scoring, goal programming or the analytic hierarchy process; cf. [21,25,38,40]) or investigate aspects such as uncertainty, robustness or the dynamics of the portfolio selection (e.g., [24,28,29]). Secondly, we also adopt ideas from articles dealing with *scheduling and staffing*. There is a vast literature on this topic; part of it emphasizes the scheduling view, as the papers on the *resource-constrained project scheduling problem* [26], whereas in other work the staff assignment view prevails. Typically, the latter group also addresses the question of different skills or *competencies* of employees [1,13,45]. (For competency modelling, cf. [30,35].) Since

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competencies are not fixed over time, but may be trained on the one hand and may also deteriorate if not kept “up to date” on the other hand, the question of *competence development* is raised. Some articles explicitly treat competence development in a quantitative model: these works (e.g., [4,8,20,33,39,44]) form the third strand of literature relevant for our investigation. None of these articles, however, integrates portfolio selection, staff assignment and learning in a holistic model.

In a previous paper [18], we studied a model combining portfolio analysis, staff assignment and learning within a *single-objective* framework. Considering the *multi-objective* nature of practical project management, the present paper is concerned with identifying (an approximation to) the set of Pareto-optimal solutions and thus preparing the ground for an interactive system that supports decision makers in identifying their individually “best” solution. For the computational solution, we will compare two multi-objective metaheuristics techniques, namely the Nondominated Sorting Genetic Algorithm II (NSGA-II [9]) and Pareto ant colony optimization (P-ACO [10]).

The remainder of this paper is organized as follows: Section 2 provides the formulation of the multi-objective mathematical programming model and provides linear asymptotic approximations. Section 3 decomposes the problem into two subproblems and introduces solution procedures. Next, Section 4 describes experiments with synthetically generated test cases to illustrate the performance of our solution techniques. Finally, a real-life application in the Electronic Commerce Competence Center Austria is provided in Section 5, while conclusions, as well as an outlook for further research, are presented in Section 6.

2. Model formulation

2.1. Mathematical definition

Our multi-objective decision analysis model builds on the single-objective *Project Selection, Scheduling and Staffing with Learning* problem (short: PSSSL problem) introduced in [18]. In the sequel, the essential elements of the PSSSL are recapitulated. After that, we extend the PSSSL by the introduction of multiple objective functions, which leads to a multi-objective mixed-integer optimization problem that we abbreviate by MPSSSL.

2.1.1. Project portfolios

Suppose a set of *candidate projects* indexed by $i = 1, \dots, n$ to be given. Our aim is to provide decision support for the selection of a subset of projects, that is, a so-called *project portfolio*. The decision which candidate project is to be realized is represented by decision variables y_i ($i = 1, \dots, n$), where y_i takes the value 1 if project i is included in the portfolio, and the value 0 otherwise. Note that we only allow 0-1-decisions about projects, that is, we do not consider the possibility of funding projects only partially.

A fixed time interval consisting of T periods is considered. Periods are indexed by $t = 1, \dots, T$. (Typically, a period consists of one month.) Period t starts at time $t - 1$ and ends at time t . We restrict ourselves to a *static* version of the decision problem where it is assumed that the decision on the projects to be selected has to be made at time $t = 0$, the start time of period 1, and remains invariant until the end of the time horizon (time T). Our model provides flexibility by allowing the definition of sub-elements of projects, called *tasks*. In total, we assume $K \geq n$ tasks, indexed by $k = 1, \dots, K$. Each project can consist of one or several tasks; each task must be assigned to exactly one project. The assignment of tasks to projects is described by constant binary numbers c_{ik} , where, for each project i and task k , we set $c_{ik} = 1$ if project i contains task k , and $c_{ik} = 0$ otherwise. By the condition above, $\sum_{i=1}^n c_{ik} = 1$ for all k .

Furthermore, we assume that for each task k , the following numbers are given: (i) the *ready time* $\rho_k \in \{1, \dots, T\}$ of task k , and (ii) the *due date* $\delta_k \in \{1, \dots, T\}$ of task k . The ready time ρ_k and the due date δ_k are defined as the first and the last period, respectively, where work in task k is possible; that is, work in task k can start not earlier than at time point $\rho_k - 1$ and must be completed not later than at time point δ_k .

2.1.2. Employee allocation

Our analysis will be carried out not on the aggregated level of the entire working team, but on the individual level of employees, which is much more realistic in several aspects (concerning work assignment and competence growth) than an aggregated consideration. The *employees* are indexed by $j = 1, \dots, m$. We suppose that the set of employees is fixed during the entire planning interval $[0, T]$. In other words, extensions of the staff, terminations of employment, outsourcing etc., are not taken into account. The different fields of knowledge, education, skills etc., in which the employees can have abilities (relevant for the company) are called *competencies*. We index competencies by $r = 1, \dots, R$. The degree to which an employee j possesses a certain competency r at time t is quantified by a real (possibly also negative) value z_{jrt} which we call the *competence score*. It is assumed that by learning, z_{jrt} grows when employee j works in a task requiring competency r , and that z_{jrt} diminishes by the so-called knowledge depreciation effect when employee j is not active in competency r . Initial values z_{jr1} of the competence scores in period 1 are assumed as given. Methods for measuring competence scores will not be discussed in this paper; as to this subject, we refer to the literature on labor psychology.

The *efficiency* of employee j in competency r , denoted by γ_{jrt} , is defined as the share of work performed in one time unit by employee j on a task requiring only competency r , if the entire task takes one time unit for an employee with “perfect ability” in competency r (cf., e.g., [44]). Although it can be expected that in general, the efficiency value will be an increasing function of the competence score, the exact functional relation depends on the way the competence score has been measured. For our purposes, we assume that γ_{jrt} can be obtained from z_{jrt} by applying some (in general nonlinear) increasing function φ_r , which may depend on the specific competency r . The function φ_r maps the set of reals into the interval $[0, 1]$.¹ A viable approach to approximate φ_r is to consider a parametrized class of functions suggested by theoretical considerations, and to estimate the parameters from empirical data. In the present paper, a logistic function (cf. [8,32]) was utilized to transform the competence score to an efficiency value, i.e., we specified the function $\varphi_r(z)$ as $\varphi_r(z) = [1 + a \exp(-bz)]^{-1}$.

Task k is assumed to require an overall *effective work time* of d_{kr} in competency r ($k = 1, \dots, K$; $r = 1, \dots, R$). The effective work time d_{kr} is the time required by an employee with maximal efficiency $\gamma_{jrt} = 1$ for completing that part of the task that is related to competency r . We call this part the *work package* with index (k, r) . As the unit for work times, we take the overall maximum possible work time in one period. In this paper, the (real) numbers d_{kr} are assumed as deterministic and known in advance.

To complete the description, *capacities* of employees are to be defined. The free capacity of employee j in period t is denoted by $a_{jt} \in [0, 1]$, expressed in work time units ($j = 1, \dots, m$; $t = 1, \dots, T$).

In each period, the effective work time in work package (k, r) can be bounded by organizational or technical constraints not to exceed a value b_{kr} ($k = 1, \dots, K$; $r = 1, \dots, R$). The numbers b_{kr} (which can also take the value ∞) are assumed as given.

¹ Our model allows negative values of the competence score. This does not mean, however, that the *efficiency* in the corresponding competence can fall below zero.

To describe (i) the scheduling of the selected projects over time with respect to their required work times, ready times and due dates, and (ii) the assignment of staff to the tasks of the selected projects with special attention given to the required competencies, we introduce real-valued decision variables $x_{kirt} \in [0, 1]$, where x_{kirt} denotes the *real* time employee j works within period t in competence r of task k ($k = 1, \dots, K; j = 1, \dots, m; r = 1, \dots, R; t = 1, \dots, T$). As in the case of effective work times and capacities, time is again measured in multiples of the overall maximum possible work time in one period. The amount of effective work time contributed by real work time x_{kirt} of employee j in competency r and period t is then given as $\gamma_{jrt}x_{kirt}$. In total, the variables x_{kirt} form the (four-dimensional) *work time array* x .

2.1.3. Competence dynamics and learning

To represent *learning*, the competence score of an employee j in competency r is assumed to increase in each period where employee j has worked during an amount x of (real) time in competency r by an increment of size $\eta_r \cdot x$, where the factor η_r is a constant that can depend on r . Similarly, we assume that the competence score of an employee j in competency r is reduced by the amount β_r in each time period by knowledge depreciation. Evidently, this loss can be over-compensated by the gain achieved by activity in competency r provided that $\beta_r < \eta_r$. The parameters η_r and β_r are called the *learning rate* and the *depreciation rate* of competency r , respectively.

The competence-oriented objectives refer to the end of the planning horizon, that is, the situation in (the beginning of) period $T + 1$. To define the objective functions, suppose that vectors $w^{(\pi)} = (w_1^{(\pi)}, \dots, w_n^{(\pi)})$ of *economic benefits* from projects 1 to n are given ($\pi = 1, \dots, p$). For example, $w_i^{(1)}$ and $w_i^{(2)}$ can measure profit contribution and turnover contribution, respectively, achieved from project i ($i = 1, \dots, n$).

Furthermore, suppose that for $\kappa = 1, \dots, q$, vectors $v^{(\kappa)} = (v_1^{(\kappa)}, \dots, v_R^{(\kappa)})$ of *competence weights* for competencies 1 to R are given. Each vector $v^{(\kappa)}$ represents a (desired) *competence profile*. Different competence profiles can reflect different strategic viewpoints or aims of different stakeholders. (Allowing multiple weight vectors is in agreement with classical approaches in decision analysis, cf. [2,43].) The κ th strategy weight $v_r^{(\kappa)}$ quantifies the importance of competency r with respect to competence profile κ . In the multi-objective decision approach proposed in this paper, it is not necessary that the competence weights $v_r^{(\kappa)}$ are scaled in a specific way in relation to the economic benefits $w_i^{(\pi)}$.

2.1.4. Mathematical programming formulation

For the arrays $x = (x_{kirt})$ and $y = (y_i)$ of decision variables, we define two sets of objective functions:

$$f^{(\pi)}(y) = \sum_{i=1}^n w_i^{(\pi)} y_i \quad (\pi = 1, \dots, p) \quad (1)$$

and

$$g^{(\kappa)}(x) = \sum_{r=1}^R v_r^{(\kappa)} \sum_{j=1}^m (\gamma_{jrt,T+1} - \gamma_{jrt}) \quad (\kappa = 1, \dots, q). \quad (2)$$

The first set (1) of objective functions represents the economic benefits from the selected projects. The objective function $f^{(\pi)}(y)$ in this set measures the economic benefit according to the values $w_i^{(\pi)}$ assigned to the single projects. The second set (2) of objective functions represents the competence benefits obtained from the increments of the efficiencies γ_{jrt} over the planning horizon. The objective function $g^{(\kappa)}(x)$ in this set measures the total increment of weighted efficiencies, cumulated over employees, where the efficiency value corresponding to competency r is weighted by the

importance value $v_r^{(\kappa)}$. Contrary to (1), the objective functions (2) do not depend on the portfolio decision vector y in a direct way, but only indirectly through the work time array x .

The problem MPSSSL is then defined as follows:

$$(f^{(1)}(y), \dots, f^{(p)}(y), g^{(1)}(x), \dots, g^{(q)}(x)) \rightarrow \max \quad (3)$$

subject to (1), (2) and

$$\gamma_{jrt} = \varphi_r(z_{jrt}) \quad (4)$$

$$z_{jrt} = z_{jrt-1} - \beta_r(t-1) + \eta_r \sum_{k=1}^K \sum_{s=1}^{t-1} x_{kirs} \quad \forall j, r, t \quad (5)$$

$$\sum_{k=1}^K \sum_{r=1}^R x_{kirt} \leq a_{jt} \quad \forall j, t \quad (6)$$

$$\sum_{t=\rho_k}^{\delta_k} \sum_{j=1}^m \gamma_{jrt} x_{kirt} = d_{kr} \sum_{i=1}^n c_{ik} y_i \quad \forall k, r \quad (7)$$

$$um_{j=1}^m \gamma_{jrt} x_{kirt} \leq b_{kr} \quad \forall k, r, t \quad (8)$$

$$(t - \rho_k) x_{kirt} \geq 0 \text{ and } (\delta_k - t) x_{kirt} \geq 0 \quad \forall k, j, r, t \quad (9)$$

$$x_{kirt} \geq 0 \quad \forall k, j, r, t, \quad y_i \in \{0, 1\} \quad \forall i. \quad (10)$$

Constraints (4) specify the dependence of the efficiency values γ_{jrt} on the competence scores z_{jrt} . Constraints (5) represent the evolution of the competence scores by knowledge depreciation and by learning. Note that we assume that the competence score remains fixed *during* a period, which is only a valid approximation if the period length is chosen as comparably short. Constraints (6) bound the invested real work time of each employee by her or his capacity limit. Constraints (7) ensure that the real work time of each employee in a competency r within a given task k , multiplied by her or his efficiency (which gives the effective work time), and cumulated over all employees and over the runtime of the task, must yield the overall required effective work time d_{kr} for work package (k, r) , if the project to which task k belongs is selected in the portfolio, and zero otherwise. Constraints (8) bound the effective work times in each work package by the maximum allowed amount per period. Constraints (9) ensure that before the ready time and after the due date of a task, no work is spent to this task. Constraints (10) restrict the decision variables to their allowed ranges.²

As it can be seen, the formulation above allows it to distinguish different categories of economic gains and to represent them as different objectives. Similarly, different “strategic lines” for competence development, each represented by a weight vector $v^{(\kappa)}$, can be taken into account in the form of separate objectives.

The elicitation of the weights $v_r^{(\kappa)}$ may be supported by a variety of methods (for an overview, cf. [5]). A natural choice would be to rely on the preference comparison methods of Multiattribute Utility Theory (MAUT), applying an additive model which is generally quite robust [6] and is consistent with our linear problem formulation. Also applying the analytical hierarchy process (for an example, cf. [2]) may be a promising approach. When eliciting the weights, the assessment of each vector $v^{(\kappa)}$ can be based on a separate stakeholder group.

Even in the special case where the functions φ_r are linear, the MPSSSL problem is a nonlinear multi-objective problem: The variables γ_{jrt} , which depend on the decision variables x_{kirt} by (4) and (5), are multiplied by the variables x_{kirt} in (7).

2.1.5. Pareto-optimal solutions

Because of the multi-objective nature of our problem formulation, the decision maker cannot be provided with a single “opti-

² By a very small extension of the model, also *formal training* of employees, e.g., courses where specific skills are acquired, could be represented. We omit the details for the sake of brevity.

mal” solution. Instead, we determine (an approximation to) the set of *Pareto-optimal solutions*. Let us repeat some basic definitions. The decision variable, that is, the “solution” to the problem, is denoted by u in the definitions below. In the case of the MPSSSL problem, u is given by the pair (y, x) , where y is the (binary) project portfolio vector, and x is the (real-valued) work time array. The objective functions (to be maximized) are written as $\Psi_1, \dots, \Psi_D$ below; in the MPSSSL case, these D objective functions consist of the two groups $f^{(1)}, \dots, f^{(p)}$ and $g^{(1)}, \dots, g^{(q)}$, such that $D = p + q$. A solution u dominates a solution u' if $\Psi_\vartheta(u) \geq \Psi_\vartheta(u')$ for all $\vartheta = 1, \dots, D$, and $\Psi_\vartheta(u) > \Psi_\vartheta(u')$ for at least one ϑ . A solution u^* is called *Pareto-optimal*, if there is no feasible solution that dominates u^* .

As seen from [18], already the single-objective version PSSSL of the MPSSSL problem is hard to solve, which is not surprising in view of the nonlinear and mixed-integer problem characteristics. For this reason, it cannot be expected that real-world instances of the MPSSSL problem can be solved exactly within reasonable computation time. Instead, we shall propose the application of multi-objective metaheuristics in order to obtain suitable approximations to the set of Pareto-optimal solutions.

2.2. Linear asymptotic approximation

Usually, both learning rates η_r and depreciation rates β_r are small compared to unity.³ This enables an asymptotic approximation that transforms the originally nonlinear problem formulation (3)–(10) into a linear one, which is somewhat easier to solve. Even in cases of not very small rates η_r and β_r , the solution obtained by the asymptotic linearization might be used as an initial solution for a local search where the decision is fine-tuned in the nonlinear context. For the single-objective case, a corresponding asymptotic approximation has been presented in [18]. The multi-objective situation can be treated similarly; thus, we keep our presentation short, focus on the differences and refer the reader to [18] for technical details. Mathematically, the assumption of small learning rates η_r and small depreciation rates β_r can be represented by setting

$$\eta_r = \bar{\eta}_r \cdot \epsilon \quad \text{and} \quad \beta_r = \bar{\beta}_r \cdot \epsilon, \quad (11)$$

where $\bar{\eta}_r$ and $\bar{\beta}_r$ are constants, and $\epsilon \ll 1$. First of all, note that the objective functions $f^{(\pi)}(y)$ in (1) do not depend on ϵ (and are already linear), so nothing has to be done there. By combining (4) and (5) to a single equation and inserting (11), we obtain $\gamma_{jrt} = \gamma_{jrt}(\epsilon) = \varphi_r(z_{jr1} + \epsilon h_{jrt})$ where

$$h_{jrt} = -\bar{\beta}_r(t-1) + \bar{\eta}_r \sum_{k=1}^K \sum_{s=1}^{t-1} x_{kjsr}.$$

Taylor expansion at $\epsilon = 0$ and omission of terms of order $O(\epsilon^2)$ yields

$$\gamma_{jrt} \sim \varphi_r(z_{jr1}) + h_{jrt} \cdot \varphi'_r(z_{jr1}) \cdot \epsilon. \quad (12)$$

As a consequence, the objective function $g^{(k)}(x)$ in (2) can be approximated by

$$\begin{aligned} & \sum_{r=1}^R v_r^{(k)} \sum_{j=1}^m (h_{j,r,T+1} - h_{j,r1}) \varphi'_r(z_{jr1}) \epsilon \\ &= \epsilon \cdot \sum_{r=1}^R v_r^{(k)} \sum_{j=1}^m \varphi'_r(z_{jr1}) \left\{ -\bar{\beta}_r T + \bar{\eta}_r \sum_{k=1}^K \sum_{s=1}^T x_{kjsr} \right\}. \end{aligned}$$

It is easy to see that by transforming an objective function in a multi-objective optimization problem by an increasing transformation function, the set of Pareto-optimal solutions does not change, since the dominance relations between solutions remain invariant. Ob-

serve that ϵ , $v_r^{(k)}$, $\varphi'_r(z_{jr1})$ and $\bar{\beta}_r T$ do not depend on the decision. Therefore, instead of maximizing the expression approximating $g^{(k)}(x)$ above, one can also maximize the expression obtained by dividing the original expression by $\epsilon > 0$ and by adding then the constant $\sum_{r=1}^R v_r^{(k)} \sum_{j=1}^m \varphi'_r(z_{jr1}) \cdot \bar{\beta}_r T$. This yields the following transform of the approximated objective function:

$$\bar{g}^{(k)}(x) = \sum_{r=1}^R v_r^{(k)} \bar{\eta}_r \sum_{j=1}^m \varphi'_r(z_{jr1}) \sum_{k=1}^K \sum_{s=1}^T x_{kjsr}. \quad (13)$$

Let us now consider the constraints. We have already dealt with (4) and (5). Constraint (6) is linear. There remain constraints (7) and (8), containing the efficiencies γ_{jrt} . From (12) we see that a first order-approximation for $\sum_{j=1}^m \gamma_{jrt} x_{kjsr}$ is given by

$$\sum_{j=1}^m \gamma_{jrt} x_{kjsr} \sim \sum_{j=1}^m \varphi_r(z_{jr1}) x_{kjsr} = \sum_{j=1}^m \gamma_{jr1} x_{kjsr}. \quad (14)$$

Considering also the $O(\epsilon)$ approximation term from (12) in (7) or (8) would cause an influence of order $O(\epsilon)$ on the variables x_{kjsr} , which would add a correction term to the objective function (13) that is already negligible compared to the main term. Hence, also the approximated constraints are linear in the decision variables x_{kjsr} .

3. Solution techniques

The MPSSSL problem admits a natural decomposition into two subproblems: The *master problem* consists in the portfolio selection, i.e., in the choice of the binary vector y . This is a discrete multi-objective optimization problem with the set $\{0, 1\}^n$ of binary vectors of length n as the search space, and $p + q$ objectives. If a special solution y is considered, the values of these $p + q$ objectives are (in general) not yet completely determined; instead, the solution of the slave problem described below assigns to the given y a (possibly empty) set of solutions corresponding to a set of points in the objective space.

The *slave problem* consists in the scheduling-and-staff assignment decision, i.e., in the choice of the work time array x , given a fixed portfolio y . Contrary to the master problem, the slave problem is a *continuous* multi-objective optimization problem. Its search space consists of the feasible work time arrays x for the given fixed portfolio y (this search space can also be empty). Since for fixed y , the first p objective functions in (3) become fixed, the slave problem has actually only q objectives. If the linear approximations of Section 2.2 are applied, the slave problem reduces to a multi-objective *Linear Program* (LP) given by $(\bar{g}^{(1)}(x), \dots, \bar{g}^{(q)}(x)) \rightarrow \max$ with $\bar{g}^{(k)}(x)$ defined by (13), subject to the constraints (6)–(10). We shall focus on the special case $q = 1$ where the slave problem has actually only one objective, such that it consists in the solution of an ordinary (single-objective) LP.⁴

We apply two multi-objective metaheuristics for the approximate solution of the master problem: the Nondominated Sorting Genetic Algorithm II (NSGA-II) by Deb et al. [9], and the Pareto ant colony (P-ACO) algorithm by Doerner et al. [10]. The solution x returned by the procedure for the slave problem to the given portfolio y is either unique or empty. If a non-empty solution $x = x(y)$ has been obtained for some y , the full vector of objective function values can be determined by the master procedure. Otherwise, the given portfolio y does not admit a feasible work time array x . Thus, any multi-objective metaheuristic can be applied in the

³ Note that the typical length of a period is one month. Over the entire planning horizon T , small increments/decrements by learning may nevertheless cumulate to crucial differences. This does not hold anymore for terms that are small of second-order, i.e., the $O(\epsilon^2)$ terms in the expansion below.

⁴ The case $q > 1$ is considerably harder to treat computationally, since in this case, Pareto-optimal solutions are of mixed-integer type. Note that to each portfolio y that occurs in the set of Pareto-optimal solutions, the solution of the slave problem consists of a composition of $(q - 1)$ -dimensional facets in the q -dimensional space. These solutions can be determined (cf. [3,36]), but their composition over all possible y is very difficult. A more viable technique consists in solving also the slave problem only *heuristically*, approximating the continuous Pareto front for fixed y by a discrete (finite) number of points.

master procedure in a standard way, with the only exception that one has to take care of the case where to some y , no feasible x is found.

3.1. NSGA-II

NSGA-II is a multi-objective Genetic Algorithm (GA) searching for an approximation of the Pareto-optimal set by the successive computation of a series of generations of solutions. In each iteration, the algorithm derives a new generation from the current one by a mechanism including three particular algorithmic components: (i) fast-non-dominated-sort, (ii) crowding-distance-assignment, and (iii) the genetic operators crossover and mutation. We cope with the situation where no feasible x is found by a *repair mechanism*: Randomly selected projects are removed from portfolio y by setting the corresponding genes y_i to zero, until feasibility is achieved. Note that in our framework, by a suitable reduction of the set of projects to be included into the portfolio, it is always possible to reach a portfolio for which a feasible work time array x exists (in the extreme case, this is the empty portfolio).

3.2. P-ACO

The P-ACO algorithm generalizes the ant colony optimization (ACO) metaheuristic (see [11]) for single-objective problems to the case of several objective functions, determining approximations to the set of Pareto-efficient solutions. ACO is a nature-inspired metaheuristic where solutions are constructed randomly and step-by-step. Construction steps that have turned out as part of good solutions in previous iterations of the construction process are favored by means of “pheromone values” during the current iteration. In our application, where a solution consists of a binary vector $(y_1, \dots, y_n)$, we define a construction step as the assignment of a value 0 or 1 to one of the binary variables y_i . (This corresponds to the use of the so-called *chain* construction graph introduced in [15]). Values are assigned from left to right, i.e., for bit $1, 2, \dots, n$. In order to guarantee feasibility of the obtained solution, the following problem-dependent rule is used for our application: If up to now, the first $i - 1$ decision variables have been set to the values $y_1, \dots, y_{i-1}$, then the next variable y_i is only allowed to be set to the value 1 if the project portfolio $(y_1, \dots, y_i, 0, \dots, 0)$ has a feasible work time array x . Whether this is the case or not is judged by the slave procedure. If it is the case, both values 0 and 1 are feasible for the current variable y_i ; otherwise, only the value 0 is feasible.

P-ACO extends ACO (i) by an additional outer iteration in which random weights for each objective function are chosen, (ii) by checks whether a newly found solution is non-dominated by candidate solutions in a current solution set and vice versa, and (iii) by an objective-specific pheromone handling mechanism. For the selection of promising solutions in each step of the inner iteration, P-ACO needs a scalarizing function. We applied the weighted Chebyshev distance function to an ideal point of the problem, i.e., a point each component of which is an upper bound of the corresponding objective function values (cf. [34]).

4. Experiments with synthetic test cases

In the synthetically generated test cases,⁵ we chose only one economic and one competence objective function, i.e., we set $p = 1$

and $q = 1$. We varied three factors that might influence the results of the used metaheuristics.

- (1) *Instance size*: We generated instances with $n = 12$ projects (“small instances”) and with $n = 18$ projects (“large instances”).
- (2) *Tightness of capacity constraints*: We generated different *project sizes*, where the size ζ_i of project i is defined as the overall effective work time required by project i . First, an average project size $\bar{\zeta}$ was calculated as $\bar{\zeta} = Tm\mu/n$, where T is the number of periods, m is the number of employees, and n is the number of projects. The multiplier μ was used to define whether the capacity constraints are tight or loose: (i) tight: $\mu = 1.25$, (ii) loose: $\mu = 1.00$. Then for each project $i = 1, \dots, n$, a random number ξ_i was drawn from a uniform distribution on $[0, 1]$, and ζ_i was determined as $\zeta_i = (\bar{\zeta}n\xi_i)/(\sum_{j=1}^n \xi_j)$. Finally, the project sizes ζ_i were split randomly into the effective work times d_{kr} required by competency r in task k assigned to project i . In the synthetically generated test instances, we identified projects and tasks, i.e., we let each project consist of only one task. Economic benefits $w_i^{(1)}$ were determined as $\zeta_i \cdot (\text{const} + \text{noise}_i)$, where noise_i is a random noise term with uniform distribution and mean zero.
- (3) *Distribution of the competence weights*: We chose $R = 20$ competencies and investigated two distribution models: (i) *Random*: To each competence, a weight $v_r^{(k)}$ was assigned by drawing from a uniform distribution on $[0, 1]$. (ii) *Counter-economic*: In order to study the tradeoff between economic and competence benefits, we used the following rule to generate the competence weights. First, the projects were split into two groups with comparably low resp. high economic benefit. A competence weight $v_r^{(1)}$ drawn from a uniform distribution on $[0, 1]$ was assigned to competencies that were strongly required by the low-benefit group. The competence weights of other competencies were set to zero. The possible combinations of levels for these three factors yield eight different types of test cases. For each type, ten independent test cases were generated randomly, leading us to obtain 80 test cases in total. The other parameters were generated as follows: (a) *Ready times and due dates*: The values ρ_k and δ_k for each task k were determined by drawing two uniformly distributed random numbers $\xi^{(1)}$ and $\xi^{(2)}$ from $[0, 1]$, multiplying them by T , and rounding to integers. The smaller resulting number determines the ready time, the larger determines the due date. (b) *Efficiencies*: Initial efficiency values γ_{jr1} for each employee j and each competency r in period 1 were drawn from a uniform distribution on $[0, 1]$. The initial competence scores z_{jr1} were obtained by applying the inverse function ϕ_r^{-1} to ϕ_r . (c) *Capacities*: The capacities a_{jt} were determined based on a uniformly distributed random variable on $[0.5, 1]$. (d) *Rates*: Learning rates η_r and depreciation rates β_r for each competency r were drawn from uniform distributions, using different scales with a factor of 800 between the maximum values of η_r and β_r , respectively. For each of the 80 test cases, we performed ten runs of our metaheuristics with different seeds. Our experiments were performed on a PC with 3.2 GHz, using C++ and ILOG CPLEX version 11.0. For the test cases of small instance size ($n = 12$), we compared the results of the metaheuristics with a procedure where on the master problem level, the multi-objective metaheuristic was replaced by complete enumeration (CE) and determination of the exact Pareto front. These runs required about 4.1 h per test case. We gave the metaheuristics 5% of the runtime of the CE runs, i.e., each metaheuristic was given 12 min computation time. The test

⁵ In default of comparable integrated models, we cannot test our procedures on available benchmarks. There are test instances for special parts of our model. E.g., in [31], test cases for multi-objective project selection problems are obtained using Steuer's [37] ADBASE code. However, for our purposes, information on competence scores of employees, competence requirements of projects, learning rates etc., would have to be filled in, such that we found it more appropriate to generate our test cases from scratch.

cases with large instance size ($n = 18$) were given 27 min computation time.

For the evaluation of the quality of the solution sets delivered by the multi-objective metaheuristics, we chose three measures:

- (1) The *hypervolume* measure (see [46–48]) which is defined as the volume of the objective space dominated by an approximation set. Its specification requires the definition of a reference point (worst case point). In our case, the zero vector can be used as the reference point, provided that competence gains are not measured with respect to the initial state as in Eq. (2), but rather with respect to the state resulting (by the choice of the empty portfolio) from maximum competence loss during the planning horizon. The higher the hypervolume, the better the approximation set.
- (2) The *spacing* measure (measure Q_5 in [23]) which quantifies the quality of the distribution of the proposed efficient solutions in objective space. Its value is the lower, the better this distribution is.
- (3) The *coverage* measure (measure Q_6 in [23]). It is a binary indicator; coverage (A, B) gives the share of solutions in set B that are dominated by the solutions in set A . In our experimental evaluation where 10 runs were performed for each test case and each metaheuristic, we computed the average $co(1)$ of all 100 coverage values coverage (A_i, B_j) for $i, j = 1, \dots, 10$ and the average $co(2)$ of all 100 coverage values coverage (B_i, A_j) for $i, j = 1, \dots, 10$, where A_i and B_i denote the solution set provided by the i th run of the first and of the second considered algorithm, respectively. If $co(1) > co(2)$, the first algorithm is better than the second one, and vice versa.

To obtain comparable magnitudes of values for the two objective functions, we rescaled both objective function values to the interval $[0, 1]$ by dividing the first objective function value $f^{(1)}(y)$ by the maximum value observed (for the considered instance) during any of the test runs of any algorithm for any portfolio y , and analogously for the second objective function.

Tables 1 and 2 show the experimental results for the 80 test instances. For the comparison between P-ACO and NSGA-II with respect to hypervolume and spacing measure, a two-sided Mann–Whitney test was used to judge statistical significance of superiority results. Significantly superior entries were marked by stars. For the coverage measure, no significance tests were performed because the condition of independent basic variables is not satisfied for the indicated aggregations. In the case of the small test instances (Table 1), we also present the hypervolumes of the exact solutions, obtained by solving the master problem by CE.

From Table 1, it can be seen that for the small test instances, P-ACO and NSGA-II are comparable in their performance with respect to hypervolume and spacing measure, but NSGA-II is better than P-ACO with respect to the coverage measure in 35 of 40 cases. For the large synthetic test instances, Table 2 shows that NSGA-II clearly outperforms P-ACO. For the small instances where exact solutions are known, we see that in most cases, the hypervolume values achieved by the metaheuristics deviate from those of the exact solutions by less than 10%.

An interesting question is whether there is any pattern of the hypervolume lost between the solutions proposed by the metaheuristics and the exact Pareto front. This can be visualized by plotting the *attainment function* (see [46–48]). In Fig. 1, this is illustrated for a selected (small) test instance. The 50% attainment function of each of the two heuristic algorithms (i.e., the border of the area of points in the plane dominated by the proposed solution sets of

Table 1

Mean values of the hypervolume, the spacing measure and the coverage measure over 10 runs for the small synthetic test cases.

T.c.	Exact		P-ACO		NSGA-II		
	hv	hv	sp	co	hv	sp	co
1	0.9339	0.8578	0.0524	0.1703	0.8394	0.0505	0.5409
2	0.9502	0.8981	0.0653	0.2856	0.9044	0.0708	0.5085
3	0.9293	0.8509	0.0576	0.1746	0.8720	0.0470	0.4752
4	0.9287	0.8513	0.0740	0.2225	0.8687	0.0652	0.4850
5	0.9497	0.8829	0.0611	0.2402	0.8901	0.0493	0.5320
6	0.9546	0.8560	0.0493	0.1291	0.9057**	0.0518	0.7828
7	0.9149	0.8372	0.0804	0.3083	0.8426	0.0921	0.4835
8	0.9214	0.8410	0.0595	0.2514	0.8379	0.0462	0.5060
9	0.8979	0.8217*	0.0515	0.2442	0.7679	0.0683	0.3888
10	0.9116	0.8132	0.0631	0.1688	0.8357	0.0566	0.5884
11	0.9429	0.8404	0.0460**	0.2433	0.8230	0.0816	0.4794
12	0.9391	0.8956**	0.0603	0.2987	0.8462	0.1073	0.2238
13	0.9220	0.8629**	0.0602	0.3609	0.8172	0.0871	0.2882
14	0.9680	0.8785	0.0541	0.1746	0.8783	0.0668	0.5643
15	0.9470	0.8710**	0.0625	0.1707	0.8080	0.0624	0.3638
16	0.9580	0.8659	0.0541	0.2355	0.8402	0.0723	0.3449
17	0.9335	0.8523*	0.0542	0.2478	0.8224	0.0821	0.3389
18	0.9355	0.8337	0.0616	0.1867	0.8342	0.0835	0.3912
19	0.9366	0.8510*	0.0572	0.2819	0.8236	0.0475	0.4091
20	0.9585	0.8696	0.0598	0.2649	0.8668	0.0785	0.4500
21	0.9652	0.8995	0.0532	0.3701	0.8927	0.0510	0.4287
22	0.9245	0.8219	0.0721	0.1652	0.8528**	0.0506*	0.6234
23	0.9370	0.8403	0.0717	0.2039	0.8676	0.0608	0.5854
24	0.9396	0.8735*	0.0457	0.6572	0.8354	0.0613	0.2380
25	0.9411	0.8745	0.0634	0.3852	0.8710	0.0581	0.3733
26	0.9261	0.8442	0.0585	0.2714	0.8537	0.0351*	0.4750
27	0.9386	0.8454	0.0455	0.3471	0.8418	0.0439	0.5374
28	0.9390	0.8440	0.0557	0.2893	0.8625	0.0630	0.4702
29	0.9166	0.8671	0.0543	0.3048	0.8556	0.0568	0.3831
30	0.9447	0.8611	0.0780	0.3175	0.8720	0.0481*	0.4923
31	0.9348	0.8437	0.0622	0.2624	0.8709**	0.0642	0.6225
32	0.9548	0.8771	0.0404	0.2565	0.8953	0.0346	0.5069
33	0.9430	0.8477	0.0575	0.1708	0.8889**	0.0435	0.6402
34	0.9269	0.8666	0.0583	0.2137	0.8755	0.0432	0.5373
35	0.9308	0.8699	0.0583	0.3067	0.8783	0.0563	0.4568
36	0.9054	0.8042	0.0615	0.1077	0.8501**	0.0445	0.7046
37	0.9126	0.8805**	0.0501	0.4391	0.8575	0.0471	0.3686
38	0.9176	0.8809	0.0442*	0.4706	0.8515*	0.0685	0.2968
39	0.9489	0.8618	0.0521	0.2844	0.8793	0.0459	0.5050
40	0.9299	0.8366	0.0712	0.3102	0.8519	0.0626	0.4781

* Indicate statistically significant superiority at level $\alpha = 0.05$.

** Indicate statistically significant superiority at level $\alpha = 0.01$.

at least 50% of the runs) is compared to the attainment function of the exact Pareto front. Obviously, the extreme solution optimizing objective 2 is approximated a *little* bit better than the extreme solution optimizing objective 1. We observed the same trend also for the other test instances.

5. Real-world application

The Electronic Commerce Competence Center (EC3) Austria, a public-private partnership institution funded by the Austrian Federal Ministry of Economic Affairs and the City of Vienna as well as by twelve private enterprises was chosen as a real-world application case. The EC3's goal is to support a fast transfer of knowledge between academic institutions and business partners. Covered research areas encompass such topics as methods of information access and information visualization, designs and mechanisms of Web-based systems, empirical business analysis by quantitative methods, and the evaluation of business ideas and models by market research.

According to the mathematical formulation of the optimization problem in Section 2, the EC3 was confronted with an extensive data requirement. (Part of the data was already necessary for our work [18]; another part, however, had to be collected newly.) A

Table 2

Mean values of the hypervolume, the spacing measure and the coverage measure over 10 runs for the large synthetic test cases. Stars as in Table 1.

T.c.	P-ACO			NSGA-II		
	hv	sp	co	hv	sp	co
1	0.8548	0.0869	0.1425	0.8634	0.0466	0.5719
2	0.8348	0.0563	0.1132	0.8569*	0.0616	0.6450
3	0.7799	0.0598	0.0838	0.8280**	0.0639	0.6543
4	0.8181	0.0657	0.0963	0.8204	0.0484	0.5563
5	0.8306	0.0678	0.0943	0.8868**	0.0660	0.6417
6	0.8347	0.0486	0.0747	0.9082**	0.0504	0.8260
7	0.8250	0.0668	0.0334	0.8750**	0.0630	0.8172
8	0.8414	0.0575	0.1064	0.8641**	0.0496	0.6022
9	0.7850	0.0700	0.0802	0.8395**	0.0629	0.7313
10	0.8298	0.1006	0.0384	0.9228**	0.0469*	0.8317
11	0.8501	0.0430	0.1342	0.8680	0.0390	0.4926
12	0.8126	0.0786	0.0964	0.8730**	0.0468*	0.6471
13	0.8245	0.0721	0.1366	0.8464	0.0515	0.5562
14	0.8068	0.0457	0.0514	0.8656**	0.0402	0.6953
15	0.8543	0.0564	0.0900	0.8952**	0.0351	0.5802
16	0.8809	0.0530	0.1253	0.8795	0.0359	0.5275
17	0.8592	0.0547	0.1091	0.8920**	0.0432	0.5880
18	0.8474	0.0473	0.1211	0.8649*	0.0511	0.5350
19	0.8625	0.0410	0.0858	0.8856**	0.0554	0.6454
20	0.8260	0.0566	0.0616	0.8662**	0.0390	0.6987
21	0.8305	0.0731	0.1626	0.8522	0.0710	0.5152
22	0.7786	0.0451	0.0908	0.7834	0.0615	0.6260
23	0.8185	0.0823	0.1156	0.8698**	0.0476*	0.7280
24	0.7662	0.0766	0.0545	0.8381**	0.0511	0.7566
25	0.7804	0.0651	0.1209	0.8728**	0.0492	0.7260
26	0.8239	0.0553	0.1670	0.8341	0.0470	0.6088
27	0.8189	0.0452	0.2303	0.8337	0.0364	0.5622
28	0.7704	0.0496	0.1032	0.8136*	0.0679	0.6922
29	0.8207	0.0674	0.1725	0.8825**	0.0525	0.7262
30	0.7939	0.0720	0.1315	0.8363	0.0571	0.6962
31	0.8557	0.0453	0.1590	0.8777**	0.0398	0.5388
32	0.8078	0.0485	0.0983	0.8645*	0.0469	0.7617
33	0.8384	0.0514	0.0741	0.8735**	0.0425	0.6560
34	0.8797	0.0432	0.2601	0.8774	0.0304*	0.5222
35	0.8393	0.0517	0.1211	0.8543	0.0404	0.5854
36	0.8099	0.0635	0.1368	0.8499**	0.0426*	0.6332
37	0.7917	0.0665	0.1108	0.8704**	0.0330*	0.7114
38	0.8237	0.0645	0.0793	0.8744**	0.0561	0.7243
39	0.7934	0.0475	0.1155	0.8371**	0.0510	0.6429
40	0.7962	0.0579	0.0779	0.8479**	0.0423	0.7024

* Indicate statistically significant superiority at level $\alpha = 0.05$.

** Indicate statistically significant superiority at level $\alpha = 0.01$.

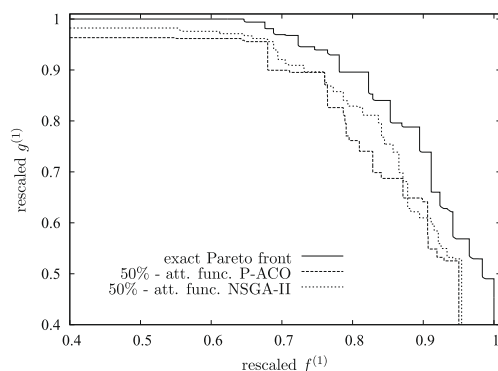


Fig. 1. Pareto front and 50% attainment functions of the P-ACO and the NSGA-II solutions for a selected synthetic test instance.

catalogue of relevant competencies as well as a competence scoring model had to be developed first. A draft for the catalogue included competencies from all four principal competence classes according to the categorization in [14], viz. personal, activity-oriented, social-communicative, and professional/methodological competence. Based on the results of an e-mail survey among

EC3's researchers, this draft underwent several adaptations. Ultimately, only the professional and methodological competencies were kept. The final version consists of $R = 80$ competencies.

Objective and subjective evidences were used as competence indicators (cf. [22]). The former distinguish formal qualifications in terms of certificates, diplomas, or publication records, and professional experience. The latter include the ratings of a researcher's competencies by him-/herself, by the scientific director, as well as by the peers in and by the head of the respective research group. The contribution of each objective evidence to each competence was specified based on background information such as curricula or journal citation indices. The subjective evidence was measured on a six-item ordinal scale according to the skill acquisition model in [12]. Objective and subjective evidence was collected by surveying via e-mail $m = 28$ employees.

From the data gathered, the competence score z_{jrt} was computed by taking the sum of the contributions of all objective evidence assigned to a researcher and adding an adjusted score obtained from the subjective competence ratings. Learning rates η_r and knowledge depreciation rates β_r were defined based on expert guesses. The knowledge depreciation rate was fixed at a rather optimistic level. Most competencies received the same learning and depreciation rates, except for several methodological competencies that were supposed to grow and diminish more slowly. The parameters a and b of the logistic function $\varphi_r(z)$ were chosen as identical for all competencies r , based on expert guesses.

For the majority of our test cases, a set of descriptions of $n = 18$ potential projects was used (data were extracted from projet plans). Three different measures of economic project benefit were provided, viz. the amount of third-party funding (ranging from approximately 200,000 Euro to no external funding at all), the overall rate of co-financing (covering the full interval from 0% to 100%), and the utility generated for business partners measured by means of an EC3-internal intellectual performance analysis (the projects included in the test data yield utility values up to 50).

Various settings of the competence weights $v_r^{(k)}$ of the competencies were devised. Five competence profiles embarking on different strategies, e.g., the orientation towards technological projects, towards data analytic and empirical projects, or towards projects in the area of mobile business, were described. Moreover, several "opportunistic" competence profiles that primarily focus on those competencies that are required in many projects, an "indifferent" competence profile with equal relative importance assigned to all competencies, as well as an "ignorant" competence profile accounting only for those competencies not required in any project were designed.

To evaluate the metaheuristics in the context of real-world scenarios, ten different test instances provided by the EC3 were used. All these instances include 18 projects, 28 employees, 80 competencies, a planning horizon of 24 months and three-objective functions in total, but they differ by the specific economic and/or the competence objective functions. Each of the two metaheuristic algorithms was given a computation time of 2 h per run for each test case.

Table 3 provides the results of the comparison between P-ACO and NSGA-II. Surprisingly, we can observe that P-ACO seems to outperform NSGA-II now. A first intuitive explanation for this phenomenon is that although we increased the runtime for the real-world test cases from 27 to 120 min, compared to the large synthetic cases, this seems not to have compensated for the increment of the number of competencies from 20 to 80 and of the number of objectives from 2 to 3. As a consequence, the given runtime in the real-instance case may still be too low for NSGA-II to deploy its full strengths. (Note that P-ACO can partially compensate for low computation time by its more "greedy" constraint handling mechanism. For larger runtime, this advantage turns into a disadvantage.) To test

Table 3

Mean values of the hypervolume, the spacing measure and the coverage measure over 10 runs for the real-world test cases. Stars as in Table 1.

T.c.	P-ACO			NSGA-II		
	hv	sp	co	hv	sp	co
1	0.6182 [*]	0.0511	0.5323	0.5693	0.0459	0.1707
2	0.8323 ^{**}	0.0964	0.7901	0.6011	0.1192	0.0993
3	0.8681 [*]	0.0813	0.7153	0.6834	0.1033	0.1277
4	0.9076 ^{**}	0.1108	0.8604	0.6922	0.1555	0.0679
5	0.8872 ^{**}	0.1543	0.5971	0.7343	0.0946	0.0710
6	0.8229 ^{**}	0.0550	0.7011	0.6503	0.1142	0.1865
7	0.9068 ^{**}	0.1164 [*]	0.8946	0.6242	0.1788	0.0294
8	0.7172	0.1692	0.2700	0.8162 [*]	0.1075	0.5945
9	0.5599	0.0787	0.5568	0.5050	0.0711	0.1950
10	0.6031	0.0444	0.4705	0.5794	0.0518	0.2361

^{*} Indicate statistically significant superiority at level $\alpha = 0.05$.

^{**} Indicate statistically significant superiority at level $\alpha = 0.01$.

this hypothesis, we further increased the runtime for the real-world instances from 2 to 4 h. In this case, P-ACO and NSGA-II performed almost equally well, perhaps even with a slight advantage for NSGA-II.

A second factor possibly disadvantaging NSGA-II in the (three-objective) real-world test cases may be the property observed in [42] that NSGA-II is especially strong in the bi-objective case. In order to study the influence of the number of objectives on the solution quality, we took the first large synthetic test case (cf. Table 2) and successively increased the number of objectives from 2 to 8. Indeed, it turned out that whereas NSGA-II dominated P-ACO for two objectives, P-ACO was significantly better than NSGA-II for the cases of 3–8 objectives, with a slightly increasing gap.

For test case 1, we show the solutions proposed by P-ACO and NSGA-II from different views: The plots in Fig. 2 present projec-

tions of the proposed Pareto-optimal solutions in objective space to the planes defined by the three possible pairs of objective functions. Objective function 1 gives partner utility, objective function 2 represents third-party funding, and objective function 3 indicates the competence gain. The objective function values have been normalized by mapping the obtained range of values to the interval [0,1].

Finally, in order to test whether our approach is still computationally feasible when the number of projects is increased, we extended the standard test instances just described by 22 additional candidate project descriptions. This produced a test case with a total number of 40 projects. P-ACO and NSGA-II performed almost equally well for this test instance, with a hypervolume of 0.2503 for P-ACO and of 0.2394 for NSGA-II.

6. Conclusions

We have developed a multi-objective model for project portfolio selection with respect to both economic and competence-oriented goals, where the latter are expressed by competency gains along certain desirable profiles. The model not only takes into account employees' different skill sets, but also learning and knowledge depreciation effects. The assignment of available personnel to work packages of the selected projects over time, i.e., over the single periods of a planning interval, is dealt with as a subproblem. The overall problem can be approximated by a *linear* (mixed-integer) multi-objective program, which allows an exact solution of the subproblem using an LP solver.

The master problem is a “rich” discrete portfolio optimization problem for which we propose to apply a multi-objective metaheuristic technique. Two metaheuristics for the described purpose have been investigated in this paper: the NSGA-II algorithm that builds on the Genetic Algorithms paradigm, and the P-ACO algorithm that makes use of the ant colony optimization approach.

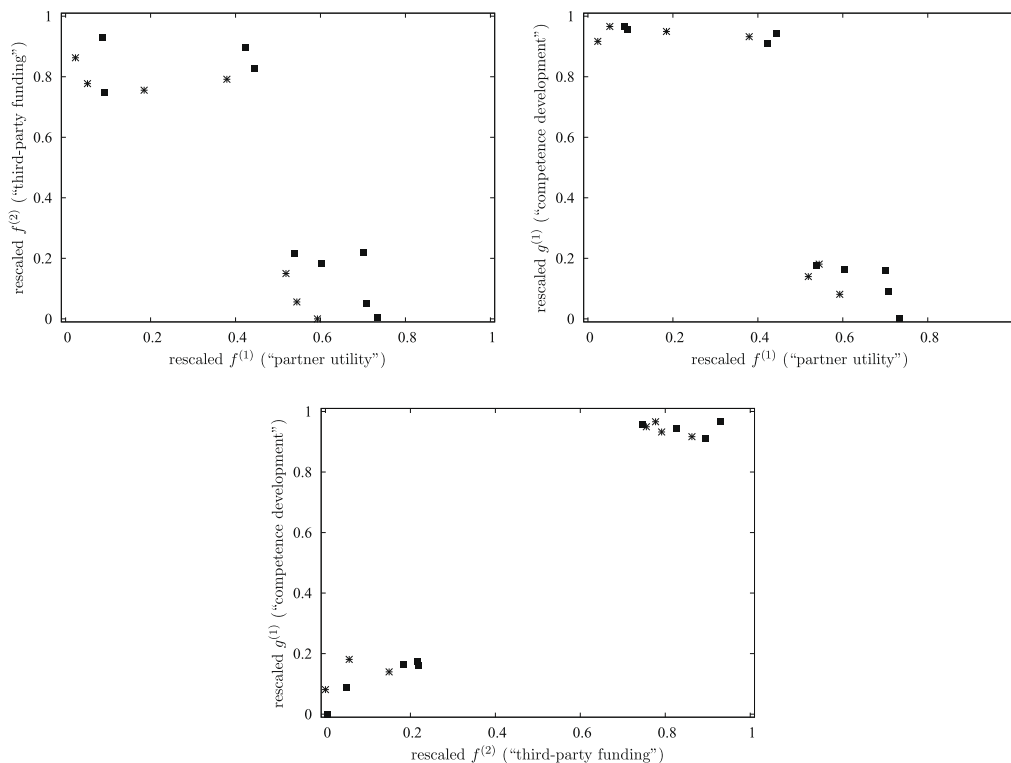


Fig. 2. Proposed Pareto-optimal solutions for real-world test case 1, projected in objective space to the three planes given by two of the axes. Filled squares: P-ACO. Crosses: NSGA-II.

We tested our proposed methods on two sets of test instances: randomly generated synthetic test cases of different size and type, as well as a real-world application delivered by the E-Commerce Competence Center Austria. In our synthetic test instances, the NSGA-II approach outperformed P-ACO. For the real-world instances, a slight superiority of P-ACO and of NSGA-II in the case of lower and of higher invested computation times, respectively, could be stated. Both techniques provided reasonable solutions from a practical point of view.

Let us outline 6 topics where future research will be particularly helpful. First, collecting data for determining the model parameters has proved as a rather time-consuming task. Suitable tools and integrated semi-automatized systems should be developed to support this process. Moreover, also methodological questions concerning the estimation of competence scores and the relation of such scores to efficiencies deserve further attention.

Secondly, the model in Section 2 does not involve precedence relations between tasks or projects. For the single-objective version of the model, it has been shown in [18] that the addition of precedence relations between tasks of a project does not violate the LP structure of the linearized problem. The same holds for our multi-objective problem formulation, and probably this observation can also be generalized to precedence relations between tasks of different projects or between projects. Therefore, the exact solutions for small instances can be determined as in Section 4 also in the case of precedence relations.

Third, whereas the model in this paper is deterministic, applications are often characterized by a large extent of uncertainty on economic or competence-oriented gains, as well as on real required (effective) work times, which need by no means to be identical to the work times estimated in advance. To adapt our approach to such a situation, the described optimization model has to be extended to a stochastic one. Some preliminary experience with a stochastic extension is already available [16,17].

Fourth, the envisaged planning time horizon in our model is rather the medium term than the long term. For long term strategic planning (sometimes already for shorter planning periods), aspects as changes of personnel or possible subcontracting or outsourcing play a role. Our model could be extended by including these aspects. E.g., external costs by subcontracting or outsourcing could be treated along the lines of the approach in [19].

Fifth, our model is static, i.e., it presupposes a fixed time horizon and a decision to be made only at the beginning of the considered period.⁶ This planning paradigm excludes adaptive policies, and it is susceptible to end-of-horizon effects. An extension of the presented approach to some form of dynamic optimization model would be as interesting as challenging.

Finally, attempts to solve instances of the considered problem by suitable *exact* methods should be made. In this context, Lagrangian duality approaches for problem decomposition (see, e.g., [27]) and hierarchical decomposition techniques for large-scale multi-objective systems, as developed in [7,41], might be explored in future research.

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⁶ In practice, it is often the case that portfolio planning is done in a rolling-horizon manner. In such a situation, our analytic approach can be applied anew each time a new decision is to be made, based on the currently available information. This “iterated static” decision process, however, is not yet a *dynamic* decision process that anticipates re-planning by changes.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at [doi:10.1016/j.ejor.2010.01.041](https://doi.org/10.1016/j.ejor.2010.01.041).

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